LITERATURE REVIEW

Attention Is All You Need

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ANALYTICAL ABSTRACT

The paper asks whether recurrence and convolution are necessary for sequence transduction, or whether a purely attention-based network can learn dependencies between positions at lower training cost. The authors propose the Transformer: stacked encoder and decoder layers built from multi-head scaled dot-product attention, position-wise feed-forward networks, residual connections, layer normalisation and fixed sinusoidal positional encodings. Evaluation uses WMT 2014 English-German and English-French translation, an ablation of architectural hyperparameters on a development set, and English constituency parsing on the Wall Street Journal treebank. The big model reaches 28.4 BLEU on English-German and 41.8 on English-French, and the base model exceeds earlier single models and ensembles on English-German at a fraction of the reported floating point cost. The review reconstructs the mechanism, derives the variance argument behind the scaling factor, checks the reported parameter and compute figures, and shows by simulation why unscaled dot products saturate the softmax. The main limitations are the absence of repeated runs or significance tests, a cost metric that ignores memory and sequence length, and the lack of a theory explaining why attention should generalise.

1 Introduction

1.1 Paper identification and classification

The paper is a computational and methodological contribution in deep learning for natural language processing, subfield neural sequence transduction and machine translation. It introduces a model architecture and evaluates it empirically. It contains no formal theorem. The version read in full is the arXiv rendering of the final revision, including the main text, Tables 1 to 4, the references and the attention visualisation figures. Footnote text was not rendered in that version, so the two footnotes (on the variance of dot products and on the floating point capacity assumed per GPU) were not read verbatim; the corresponding statements below are derived or checked independently and labelled as such. The paper PDF itself could not be downloaded in the sandbox used for this review, so it is not stored in the library.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether a network that computes representations only by letting every position weight every other position, with no recurrent state and no convolution, can match or beat recurrent and convolutional encoder-decoder models in translation quality while training faster.

WHY DOES THE QUESTION MATTER?

Recurrent models factor computation along symbol positions, so a hidden state h_t depends on h_{t-1} and training within an example cannot be parallelised. The paper argues that this sequential dependence becomes binding at long sequence lengths because memory limits batching across examples. A model whose depth of computation does not grow with sequence length can exploit parallel hardware, and a short path between distant positions should ease the learning of long-range dependencies. The question is therefore both computational (wall-clock and floating point cost) and statistical (how dependencies are learned).

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Attention was already an integral part of competitive translation systems, but almost always attached to a recurrent network. Convolutional alternatives (Extended Neural GPU, ByteNet, ConvS2S) removed recurrence but required a number of operations growing linearly or logarithmically with the distance between positions. Self-attention had been used inside recurrent or task-specific models. What was missing, in the authors' account, was a transduction model relying entirely on self-attention, and an assessment of whether such a model is competitive on a hard benchmark.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the Transformer. It investigates by training base and big variants on WMT 2014 translation tasks, comparing BLEU and an estimated training cost with published systems, varying hyperparameters one at a time on a development set, and transferring the architecture almost unchanged to constituency parsing. It finds new best BLEU scores on both translation tasks at lower estimated training cost than the strongest published systems, competitive parsing results, and a sensitivity of quality to the number of heads, the key dimension, depth, width and dropout.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can a sequence transduction model that replaces recurrent and convolutional layers by multi-head self-attention reach higher translation quality than recurrent and convolutional models while requiring less training time, and does the same architecture transfer to a structurally different task, English constituency parsing, without task-specific tuning?

2.2 Hypotheses

- H1 (paper). Attention alone suffices to model global dependencies between input and output, so recurrence and convolution are not needed.
- H2 (paper). Constant sequential depth and constant path length make the model more parallelisable and easier to train on long-range dependencies, so training cost falls for equal or better quality.
- H3 (paper). Several attention heads over lower-dimensional projections recover the representational diversity lost by averaging in a single attention operation.
- H4 (paper). Sinusoidal positional encodings let the model attend by relative position and may extrapolate to longer sequences than those seen in training.
- H5 (paper). The architecture generalises to other tasks, tested on English constituency parsing in large and small data regimes.

2.3 Existing literature and research gap

PAPER The authors position the work against recurrent encoder-decoder models with attention (Sutskever, Vinyals and Le 2014; Bahdanau, Cho and Bengio 2015; Wu et al. 2016), against convolutional models (Kalchbrenner et al. 2016; Gehring et al. 2017) and against self-attention used in reading comprehension, summarisation and sentence representation (Parikh et al. 2016; Lin et al. 2017). They state that the Transformer is, to their knowledge, the first transduction model relying entirely on self-attention.

CRITIQUE The claim of priority is narrowly worded and credible, but the novelty is one of composition. Scaled dot-product attention, residual connections, layer normalisation, dropout, label smoothing, byte-pair encoding and tied embeddings were all pre-existing; the contribution is the particular assembly and the demonstration that it works at scale. The gap addressed is empirical and architectural, not conceptual: no prior model had shown that the recurrence could be removed without loss on a flagship benchmark.

2.4 Claimed contribution

CRITIQUE The performance claim is demonstrated on two translation benchmarks against published numbers, with the caveat that comparisons mix single models and ensembles, that the cost metric is an estimate, and that there are no repeated runs. The generality claim rests on one additional task. The claim that attention is more interpretable rests on three hand-picked visualisations and is not quantified. The headline contribution (competitive quality at lower cost) is demonstrated; the explanatory claims about why it works are not.

3 Theoretical Framework and Mechanism

3.1 Scaled dot-product attention

PAPER An attention function maps a query and a set of key-value pairs to an output that is a weighted sum of the values, the weight of each value being a compatibility function of the query with its key. With queries, keys and values packed into matrices Q, K and V, the paper defines

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \tag{1}$$

Q	Matrix of queries, one row per output position, $n_q \times d_k$.
K	Matrix of keys, $n \times d_k$. Row j is the key of input position j.
V	Matrix of values, $n \times d_v$, the content that is averaged.
d_k	Dimension of queries and keys; the scaling factor $1/\sqrt{d_k}$ depends on it.
softmax	Applied row-wise, so each query obtains non-negative weights summing to one over positions.

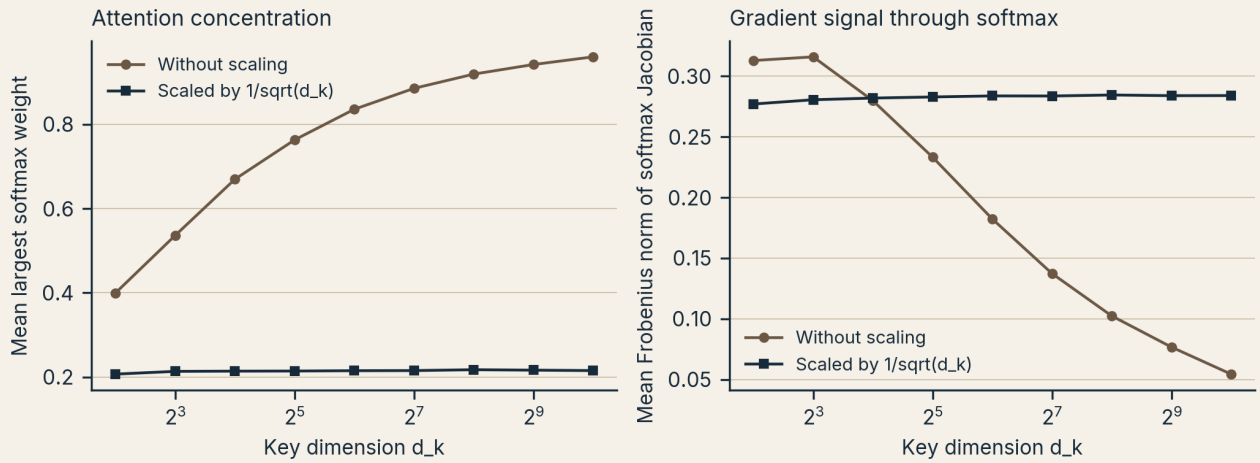
Row i of the output is $\sum_j \alpha_{ij} v_j$ with $\alpha_{ij} = \exp(q_i \cdot k_j/\sqrt{d_k}) / \sum_l \exp(q_i \cdot k_l/\sqrt{d_k})$. The output of each position is therefore a convex combination of value vectors, a content-addressed lookup in which the similarity of query and key decides how much each position contributes. The same matrix product computes all positions at once, which is the source of parallelism. In the decoder, entries of $QK^\top$ that correspond to future positions are set to $-\infty$ before the softmax, so their weights are exactly zero and the model remains autoregressive.

INFERENCE The role of the factor $\sqrt{d_k}$ can be derived. Suppose the components of a query q and a key k are independent with mean zero and variance one. Then

$$\mathbb{E}[q \cdot k] = \sum_{i=1}^{d_k} \mathbb{E}[q_i]\,\mathbb{E}[k_i] = 0, \qquad \operatorname{Var}(q \cdot k) = \sum_{i=1}^{d_k} \operatorname{Var}(q_i k_i) = d_k. \tag{2}$$

The standard deviation of an unscaled logit is thus $\sqrt{d_k}$, and dividing by $\sqrt{d_k}$ restores unit variance regardless of dimension. For $d_k = 64$, the base model value, the standard deviation is 8. When one logit exceeds the others by several units the softmax places nearly all mass on it and its Jacobian $\operatorname{diag}(\alpha) - \alpha\alpha^\top$ approaches zero, so gradients vanish. The authors state this intuition in a footnote and as a suspicion; the derivation above is the reviewer's, and Figure 1 quantifies it by simulation. The assumption of independent unit-variance components holds at initialisation only approximately and not after training.

Figure 1. **Softmax saturation with and without the scaling factor, as a function of the key dimension**



Source: generated by the reviewer. Simulation with 4,000 draws per dimension, 20 keys per query, independent standard normal components, seed 12345. Left: mean largest softmax weight. Right: mean Frobenius norm of the softmax Jacobian. Not taken from the paper.

In the simulation the variance of the unscaled logit is 64.4 at $d_k = 64$ and 1,024.1 at $d_k = 1024$, matching Equation (2). The mean largest weight over 20 keys rises from 0.40 at $d_k = 4$ to 0.84 at $d_k = 64$ and 0.96 at $d_k = 1024$, while with scaling it stays near 0.215 for every dimension. The Jacobian norm falls from 0.31 to 0.18 at $d_k = 64$ and to 0.05 at $d_k = 1024$ without scaling, and stays near 0.28 with scaling. The factor is therefore a variance normalisation that keeps the attention distribution at a fixed level of dispersion at initialisation.

3.2 Multi-head attention

PAPER Instead of one attention function over d_{model}-dimensional vectors, the model projects queries, keys and values h times with learned matrices and concatenates the results:

$$\begin{aligned} \operatorname{MultiHead}(Q, K, V) &= \operatorname{Concat}(\text{head}_1, \dots, \text{head}_h)\, W^O, \\ \text{head}_i &= \operatorname{Attention}(QW_i^Q,\, KW_i^K,\, VW_i^V). \end{aligned} \tag{3}$$

W_i^Q, W_i^K Learned projections, $d_{\text{model}} \times d_k$, for head i.
W_i^V Learned projection, $d_{\text{model}} \times d_v$.
W^O Output projection, $hd_v \times d_{\text{model}}$.
h Number of heads; the paper uses $h = 8$ with $d_k = d_v = d_{\text{model}}/h = 64$.

A single attention operation produces one weighted average, which blends all the information a position gathers into a single distribution over positions. With h heads the position can gather from h different distributions computed in different learned subspaces. Because $d_k = d_v = d_{\text{model}}/h$, the total cost is comparable to single-head attention at full dimension. The reviewer's parameter count shows the cost structure: each attention block holds $4d_{\text{model}}^2 = 1{,}048{,}576$ weights (four matrices of 512×512 when the per-head matrices are stacked), independent of h.

3.3 Position-wise feed-forward network and residual structure

PAPER Each layer also applies, to every position separately and identically,

$$\text{FFN}(x) = \max(0,\, xW_1 + b_1)\, W_2 + b_2, \tag{4}$$

with $d_{\text{model}} = 512$ and inner width $d_{ff} = 2048$. Each sub-layer, attention or feed-forward, is wrapped as $\text{LayerNorm}(x + \text{Sublayer}(x))$. Attention mixes information across positions; the feed-forward network transforms each position's representation nonlinearly, so the two alternate between communication and computation. Without the ReLU the stack would be a composition of linear maps over value vectors, with the only nonlinearity being the softmax weights. The residual connection gives each layer an identity path, which keeps the signal and the gradient from decaying through six stacked layers.

INFERENCE A reviewer's count of the base model gives $6 \times (1{,}048{,}576 + 2{,}097{,}152) = 18{,}874{,}368$ weights in the encoder (one attention block and one feed-forward network of $2 \times 512 \times 2048$ weights per layer), $6 \times (2 \times 1{,}048{,}576 + 2{,}097{,}152) = 25{,}165{,}824$ in the decoder (two attention blocks), and $37{,}000 \times 512 = 18{,}944{,}000$ in the shared embedding, a total of about 62.98 million before biases and normalisation parameters. The paper reports 65 million. The gap of about 2 million is not explained in the text read (it may reflect the vocabulary size used by the released code or components not described); this is an estimate and not an official figure.

3.4 Positional encoding

PAPER Since the network has no recurrence or convolution, position is injected by adding to the input embeddings

$$PE_{(pos,\,2i)} = \sin\!\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \qquad PE_{(pos,\,2i+1)} = \cos\!\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right). \tag{5}$$

pos Position of the token in the sequence.
i Index of the sinusoid; wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$.

The authors hypothesise that for any fixed offset k, PE_{pos+k} is a linear function of PE_{pos}. The derivation is short. Write $\omega_i = 10000^{-2i/d_{\text{model}}}$. Then

$$\begin{pmatrix} \sin(\omega_i(pos+k)) \\ \cos(\omega_i(pos+k)) \end{pmatrix} = \begin{pmatrix} \cos(\omega_i k) & \sin(\omega_i k) \\ -\sin(\omega_i k) & \cos(\omega_i k) \end{pmatrix} \begin{pmatrix} \sin(\omega_i pos) \\ \cos(\omega_i pos) \end{pmatrix}, \tag{6}$$

a rotation by the angle $\omega_i k$ that depends on the offset k and not on pos. A shift in position is therefore the same linear map at every position, which a learned query-key projection can in principle exploit to attend to a fixed relative offset. The paper tests learned embeddings instead and reports nearly identical results (Table 3, row E), so the choice is empirically inconsequential at the sequence lengths used; the extrapolation advantage is conjectured and not tested.

3.5 Complexity comparison and the argument for self-attention

PAPER Table 1 reproduces the comparison of layer types, where n is sequence length, d representation dimension, k kernel size and r the neighbourhood of restricted self-attention.

Table 1. **Per-layer complexity, sequential operations and maximum path length (reproduced from the paper)**

Layer type	Complexity per layer	Sequential operations	Maximum path length
Self-attention	$O(n^2 d)$	$O(1)$	$O(1)$
Recurrent	$O(nd^2)$	$O(n)$	$O(n)$
Convolutional	$O(knd^2)$	$O(1)$	$O(\log_k n)$
Self-attention (restricted)	$O(rnd)$	$O(1)$	$O(n/r)$

Source: Vaswani et al. (2017), Table 1 (official).

The attention cost $n^2 d$ comes from the $n \times n$ matrix $QK^\top$ with inner dimension d; the recurrent cost nd^2 comes from n matrix-vector products with a $d \times d$ matrix. Self-attention is cheaper per layer when $n < d$. **INFERENCE** For a sentence of $n = 50$ word pieces and $d = 512$, $n^2 d = 1.28 \times 10^6$ against $nd^2 = 1.31 \times 10^7$ for recurrence, a ratio of about ten in favour of attention. At $n = 2048$ the ordering reverses: $n^2 d = 2.1 \times 10^9$ against $nd^2 = 5.4 \times 10^8$. The quadratic term is the reason the paper mentions restricted attention as future work, and it does not appear in the translation experiments only because sentences are short.

3.6 Optimiser schedule

PAPER Training uses Adam with $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-9}$ and the learning rate

$$lrate = d_{\text{model}}^{-0.5} \cdot \min\left(step^{-0.5},\ step \cdot warmup^{-1.5}\right), \qquad warmup = 4000. \tag{7}$$

The first argument of the minimum decays with the inverse square root of the step, the second increases linearly; they cross at $step = warmup$. **INFERENCE** At the crossing, $lrate = 512^{-0.5} \times 4000^{-0.5} \approx 6.99 \times 10^{-4}$, and at step 100,000 (the end of base training) it has fallen to about 1.40×10^{-4}. The warm-up is needed because the Adam second-moment estimate is poorly calibrated early in training and

because post-layer-norm stacks are unstable at large initial rates; the paper does not report results without warm-up, so its necessity is not demonstrated by the authors.

4 Data and Empirical Strategy

4.1 Data

PAPER Two translation corpora and one parsing corpus are used.

Table 2. **Data and training configuration**

Task	Training data	Vocabulary	Training regime
WMT 2014 English-German	About 4.5 million sentence pairs	Shared byte-pair encoding, about 37,000 tokens	Base: 100,000 steps, 12 hours, 8 P100 GPUs. Big: 300,000 steps, 3.5 days
WMT 2014 English-French	36 million sentences	32,000 word pieces	Big model, dropout 0.1
Parsing, WSJ only	About 40,000 sentences, Penn Treebank	16,000 tokens	4-layer model, $d_{\text{model}} = 1024$
Parsing, semi-supervised	About 17 million sentences	32,000 tokens	Same architecture

Source: Vaswani et al. (2017), Sections 5 and 6 (official). Batches contain about 25,000 source and 25,000 target tokens; batches are formed by approximate sequence length.

Development results use newstest2013 and test results use newstest2014 for translation, and Section 22 and Section 23 of the Wall Street Journal for parsing development and test. The unit of observation is a sentence pair (translation) or a sentence with a parse tree (parsing). The paper gives no number of test sentences and does not describe BLEU tokenisation or any filtering.

4.2 Empirical specification and identification

EVIDENCE The design is a benchmark comparison with a controlled ablation, not a causal identification problem. The identifying variation for the claim that attention alone suffices comes from comparing the Transformer with published systems that differ in many respects at once (architecture, data processing, training budget, ensembling), so the comparison identifies the performance of the package and not the contribution of removing recurrence. The ablation in Table 3 is the only controlled variation: one hyperparameter is changed at a time on the English-German development set, holding the rest at base values. Even there each row is one training run, so the variation is not separated from seed noise. The cost estimate multiplies training time, number of GPUs and an assumed sustained single-precision capacity per GPU.

INFERENCE The reported figures are consistent with a sustained capacity of about 9.5×10^{12} operations per second per P100: $12 \times 3600 \times 8 \times 9.5 \times 10^{12} = 3.28 \times 10^{18}$ (reported 3.3×10^{18}) and $3.5 \times 24 \times 3600 \times 8 \times 9.5 \times 10^{12} = 2.30 \times 10^{19}$ (reported 2.3×10^{19}). The cost column is therefore a wall-clock proxy scaled by hardware peak, not a count of operations performed.

4.3 Data limitations and biases

- Development-set selection. Beam size, length penalty, dropout and checkpoint averaging were chosen after experimentation on the development set; selection on newstest2013 and reporting on newstest2014 is standard, but the number of configurations tried is not stated.
- Comparator heterogeneity. Table 2 mixes single models and ensembles and takes the competitors' numbers from their own papers, so differences in preprocessing, vocabulary and hyperparameter search are uncontrolled.
- Single-run reporting. No seed variance, confidence interval or significance test accompanies any BLEU figure, so differences of a few tenths of a point cannot be distinguished from noise.
- Cost proxy. Estimated floating point cost ignores memory bandwidth, utilisation differences between architectures, and inference cost.
- Domain narrowness. News text in two high-resource language pairs and one English parsing treebank; low-resource and long-document settings are not covered.

5 Results

5.1 Machine translation

EVIDENCE Table 3 reproduces the main comparison.

Table 3. **Translation quality and estimated training cost (reproduced from the paper)**

Model	BLEU EN-DE	BLEU EN-FR	FLOPs EN-DE	FLOPs EN-FR
ByteNet	23.75			
Deep-Att + PosUnk		39.2		1.0×10^{20}
GNMT + RL	24.6	39.92	2.3×10^{19}	1.4×10^{20}
ConvS2S	25.16	40.46	9.6×10^{18}	1.5×10^{20}
MoE	26.03	40.56	2.0×10^{19}	1.2×10^{20}
Deep-Att + PosUnk Ensemble		40.4		8.0×10^{20}
GNMT + RL Ensemble	26.30	41.16	1.8×10^{20}	1.1×10^{21}
ConvS2S Ensemble	26.36	41.29	7.7×10^{19}	1.2×10^{21}
Transformer (base)	27.3	38.1	3.3×10^{18}	
Transformer (big)	28.4	41.8	2.3×10^{19}	

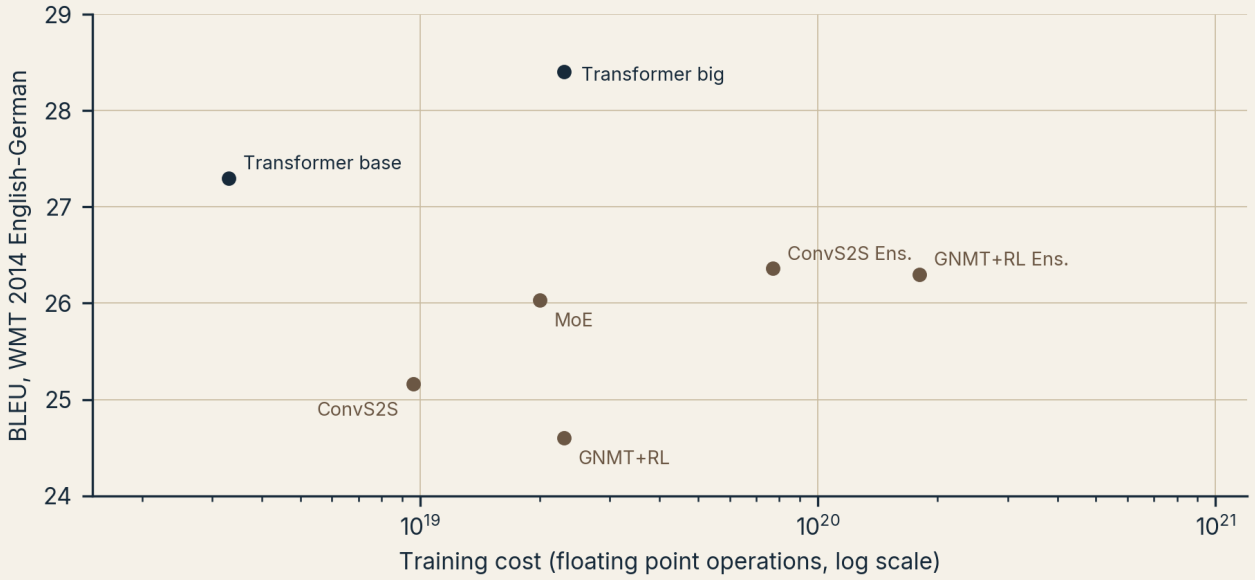
Source: Vaswani et al. (2017), Table 2 (official). Cells left empty are empty in the paper.

EVIDENCE On English-German the big model scores 28.4 BLEU, 2.04 points above the best ensemble (ConvS2S Ensemble, 26.36) and 2.4 above the best single model (MoE, 26.03). The base model, at 27.3, already exceeds every listed system including ensembles. On English-French the big model scores 41.8, 0.51 above the best listed ensemble (41.29) and 1.24 above the best single model (40.56); the base model, at 38.1, is below all the competitors listed. In cost terms, the base model needs 3.3×10^{18} operations against 9.6×10^{18} for the single ConvS2S model, a factor of 2.9, while scoring 2.14 BLEU

higher. The big model costs 2.3×10^{19}, the same as GNMT + RL on English-German, and scores 3.8 BLEU higher. On English-French the big model's cost is not printed in Table 2; the text says it is less than one quarter of the previous state of the art. The text of the version read states a French score of 41.0 while the abstract and Table 2 give 41.8; this is an internal inconsistency in the paper that the reviewer could not resolve from the available material, and Table 2 and the abstract are used here.

CRITIQUE The English-German result is the strongest evidence in the paper: a gain of more than two BLEU over every comparator with the base model alone is larger than differences normally attributed to noise, although no variance is reported. The English-French result is a smaller margin and rests on the big model. The claim of a fraction of the training cost of any competitive model holds against ensembles and against GNMT but not against the single ConvS2S model, which at 9.6×10^{18} is cheaper than the big Transformer (2.3×10^{19}) on English-German, though at a lower score.

Figure 2. **Quality against estimated training cost on WMT 2014 English-German**



Source: BLEU and training cost taken from Vaswani et al. (2017), Table 2 (official); plotted by the reviewer. Systems without a printed English-German cost (ByteNet) are omitted. Cost on a logarithmic scale.

Figure 2 plots the Pareto frontier: both Transformer points lie to the upper left of every comparator, except that ConvS2S (single) is cheaper than the big model. The vertical gap between the base model and the next best system is about 1 BLEU at lower cost, which is the substance of the efficiency claim.

5.2 Model variations

EVIDENCE Table 4 reproduces a selected set of rows of the ablation in the paper (English-German development set, newstest2013, base model otherwise, no checkpoint averaging). Perplexities are per word piece.

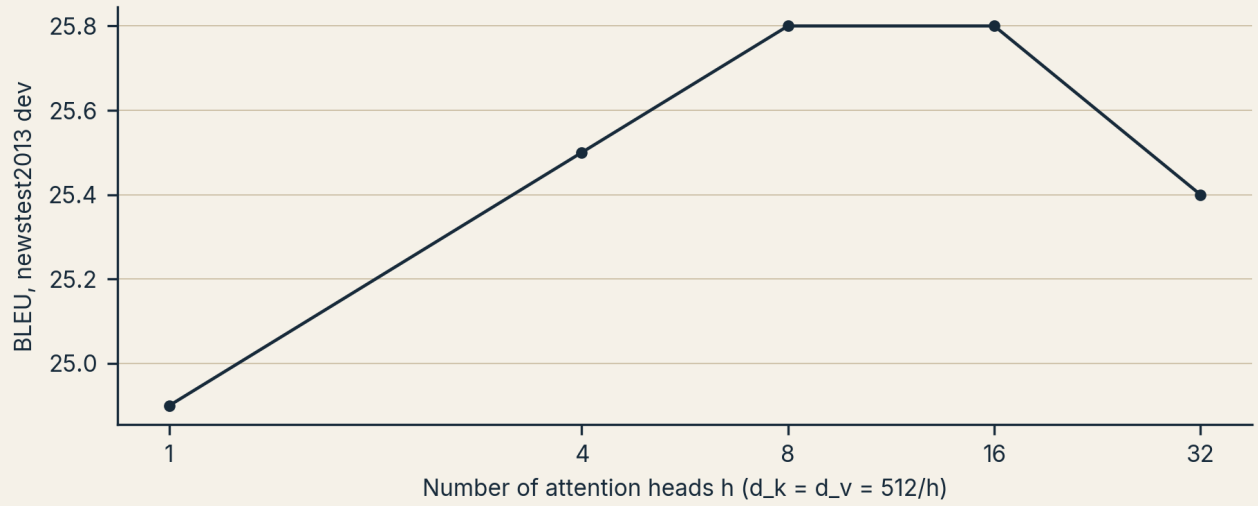
Table 4. Selected ablations on the Transformer base model (reproduced from the paper)

Variation	h	$d_k = d_v$	Other	PPL	BLEU	Params (10^6)
Base	8	64	$N = 6$, $d_{\text{model}} = 512$	4.92	25.8	65
(A) one head	1	512		5.29	24.9	
(A) four heads	4	128		5.00	25.5	
(A) sixteen heads	16	32		4.91	25.8	
(A) thirty-two heads	32	16		5.01	25.4	
(B) $d_k = 16$	8	16		5.16	25.1	58
(C) $N = 2$			$N = 2$	6.11	23.7	36
(C) $d_{\text{model}} = 1024$		128	$d_{\text{model}} = 1024$	4.66	26.0	168
(D) no dropout			$P_{drop} = 0.0$	5.77	24.6	
(E) learned positions				4.92	25.7	
Big	16		$d_{\text{model}} = 1024$, $d_{ff} = 4096$, $P_{drop} = 0.3$, 300K steps	4.33	26.4	213

Source: Vaswani et al. (2017), Table 3 (official), subset chosen by the reviewer. Blank cells repeat the base value or are blank in the paper. Development set, so not comparable with the test scores of Table 2.

EVIDENCE Single-head attention is 0.9 BLEU below the best setting and 0.9 below the base, and quality falls again at 32 heads. Reducing d_k to 16 loses 0.7 BLEU while removing only 7 million parameters. Depth matters strongly: two layers lose 2.1 BLEU. Removing dropout loses 1.2. Learned positions are within 0.1 BLEU of sinusoids.

Figure 3. Development-set BLEU as a function of the number of attention heads



Source: values taken from Vaswani et al. (2017), Table 3 rows (A) (official); plotted by the reviewer. English-German, newstest2013, single training run per point, total computation held constant.

CRITIQUE The head sweep in Figure 3 is non-monotonic with a plateau between 8 and 16, and its overall range is 0.9 BLEU. Without a seed-to-seed standard deviation, the conclusion that eight to sixteen heads are better than four or thirty-two is not statistically supported. Only the contrast between one head and the rest is large enough to be plausibly real. The ablation also changes d_k with h, so the head count and the per-head key dimension are confounded; row (B) shows d_k matters, so some of the decline at 32 heads may be a small-d_k effect.

5.3 English constituency parsing

EVIDENCE A 4-layer Transformer with $d_{\text{model}} = 1024$ obtains 91.3 F1 on WSJ Section 23 when trained on the WSJ only, 92.7 in the semi-supervised setting. Against the table of the paper, the WSJ-only score is below the recurrent neural network grammar discriminative result (91.7) and above Petrov et al. (90.4); the semi-supervised score exceeds earlier semi-supervised results (92.1) and falls short of the multi-task (93.0) and generative (93.3) systems. The paper notes that only a small number of experiments were run on dropout, learning rate and beam size using Section 22.

CRITIQUE The result supports the claim that the architecture transfers without large changes, and it matters that a model without recurrence works with 40,000 training sentences, a regime where the authors say recurrent sequence-to-sequence models had struggled. However, it is a single additional task, and the Transformer does not win in either setting against the best systems, so the paper claims surprisingly good performance, not superiority.

5.4 Interpretability of attention

PAPER Figures 3 to 5 in the appendix show attention weights for selected heads in layer 5 of 6, with heads that follow a long-distance dependency of the verb making, and heads that appear involved in anaphora resolution. CRITIQUE These are qualitative, selected examples. They show that some heads have interpretable patterns; they do not show that the model uses those patterns, since attention weights are not by themselves a causal explanation of the output, and no ablation of the highlighted heads is reported.

6 Robustness and Additional Tests

Table 5. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Gains come from one lucky configuration	Variation of heads, key size, depth, width, dropout, label smoothing on the development set	Base setting near the best of a broad range; quality falls with fewer layers, smaller keys, no dropout	Partly. The sweep shows sensitivity, but single runs with no seed variance cannot separate small effects from noise
Sinusoidal encodings are essential	Replacement by learned positional embeddings	25.7 against 25.8 BLEU	Yes for equivalence at training lengths. No for the extrapolation rationale, which is not tested
Results are specific to translation	English constituency parsing, with 40K and 17M training sentences	91.3 and 92.7 F1, competitive but not best	Partly. A second task of a different structure, but only one, and not a state-of-the-art result

Concern	Test reported	Result	Does it resolve the concern?
Comparison with prior work is unfair because of ensembling and budgets	Report of single models and ensembles and of estimated floating point cost	Base model beats all listed systems on English-German at lower cost	Partly. Costs are estimated and comparators are taken from other papers with their own tuning
Overfitting	Dropout, label smoothing, checkpoint averaging	Dropout removal costs 1.2 BLEU; label smoothing hurts perplexity but helps BLEU	Partly. Regularisers are shown to matter; no learning curves or held-out analysis beyond development BLEU
Hyperparameter selection leakage	Selection on newstest2013 and Section 22, test on newstest2014 and Section 23	Held-out test scores are reported	Largely, as the split is standard. The amount of search is not reported

Source: assessment by the reviewer from Vaswani et al. (2017), Sections 6.1 to 6.3 (tests reported by the authors; assessment generated).

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

On WMT 2014 English-German the Transformer base and big models achieve higher BLEU than every system listed in the paper, including ensembles, at an estimated training cost no larger than the strongest convolutional and recurrent competitors. Recurrence and convolution are not necessary for competitive translation quality. Depth and the key dimension matter, and removing dropout hurts.

SUPPORTED BUT CONDITIONAL

That the efficiency gain is general: it holds under the paper's cost proxy (wall-clock times hardware peak), at sentence-length sequences where $n < d$, and with comparators whose numbers come from other papers. That the architecture transfers to parsing: supported by one task, with competitive but not leading results.

SUGGESTIVE

That multi-head attention works because heads capture different representation subspaces, and that attention is more interpretable; the head ablation shows one head is worse, and the visualisations show selected patterns, but neither identifies a mechanism. That sinusoidal encodings allow extrapolation to longer sequences.

> **NOT ESTABLISHED**
>
> That attention is the cause of the improvement independent of other changes in the training recipe (label smoothing, checkpoint averaging, warm-up schedule, byte-pair vocabulary). That the gains exceed seed variation at the level of tenths of a BLEU point. That the model handles long sequences, which the paper does not test and for which the quadratic cost is a stated limitation. Any theoretical claim on expressivity or generalisation.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Parallelisability. Constant sequential depth allows training in 12 hours on eight GPUs, which matters because it makes experimentation faster.
- Short path lengths between positions, which they argue eases long-range dependency learning.
- Generalisation to parsing, including with limited data.
- Potential interpretability from attention distributions.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Clarity of ablation. Varying one component at a time with total computation held constant (the head sweep) isolates the design choice better than most contemporary architecture papers.
- Complete specification. Dimensions, optimiser, schedule, regularisation, batching and inference settings are all given, which supports replication and explains the rapid adoption.
- A reproducible efficiency metric. Even as a proxy, reporting training cost next to quality lets readers see the trade-off, and the figures are internally consistent (Section 4).
- Simple building blocks. Every component is a matrix product plus an elementwise nonlinearity, which maps directly onto accelerators, an engineering advantage that is independent of the benchmark scores.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. Every quantitative claim rests on one training run per configuration, with no seed variance, confidence intervals or significance tests, and with comparators drawn from other papers. **Why it matters.** The ablation margins (0.1 to 0.9 BLEU) and several of the Table 2 gaps are of the size at which seed-to-seed variation in neural translation is typically non-trivial, so ranking conclusions among configurations are not supported. **Potential bias.** Winner's curse: the reported base configuration and the checkpoint-averaged big model were selected after experimentation, so reported gains overstate expected performance. **How it could be tested.** Retrain base and ablation variants with at least five seeds, report means and standard deviations, and apply paired bootstrap

resampling on the test set against a re-trained recurrent and convolutional baseline under identical preprocessing and tuning budget.

OTHER WEAKNESSES

- Attribution. The Transformer changes the architecture and also introduces a new optimiser schedule, label smoothing and checkpoint averaging. Why it matters: gains may belong partly to the recipe. Test: apply the same recipe to a recurrent model.
- Cost proxy. Peak floating point capacity times wall-clock time understates the memory cost of the $n \times n$ attention matrix and says nothing about inference. Test: report measured throughput and peak memory by sequence length.
- Interpretability claims. Selected attention maps are anecdotal. Test: ablate or perturb the heads claimed to perform syntactic roles and measure the effect on output.
- Sequence length. All experiments use sentences; the $O(n^2 d)$ cost and the finite training positions leave behaviour on long inputs untested. Test: evaluate on length-stratified test sets and on document-level data.
- Internal inconsistency. The English-French score is stated as 41.8 in the abstract and table and as 41.0 in the text of the version read, which weakens confidence in careful reporting, even though the table is probably correct.
- Unexplained parameter count. The reviewer's count of the base model is about 63 million against 65 million reported; the difference is small but is not accounted for.

8 Contradictions and Edge Cases

CRITIQUE Several conditions should make the result weaker or disappear.

Long sequences. The attention matrix has n^2 entries per head. With $h = 8$ heads and $n = 4096$ at 16-bit precision the matrix alone occupies $8 \times 4096^2 \times 2 \approx 268$ megabytes per layer per example, and the compute advantage derived from Table 1 reverses once $n > d = 512$. The paper's efficiency conclusion is valid for the sentence-level regime it tests and is not a statement about long documents.

Scale of the dot product. Equation (2) assumes independent unit-variance components. If queries and keys have non-zero mean or strongly correlated components, which is common in trained networks, the variance of the logit is not d_k and the scaling may over- or under-correct; the factor is exact only at an idealised initialisation. At the other extreme, as $d_k \to 1$ the scaling is irrelevant, and the head sweep suggests very small d_k is harmful for a different reason (row B).

Softmax averaging and sharp retrieval. A convex combination of values cannot represent a precise copy of one position unless the weight on it approaches one, which requires large logits; the model therefore faces a tension between trainable gradients (moderate logits) and exact retrieval (large logits). In the simulation above, at $d_k = 64$ the unscaled model would place 0.84 of the mass on the largest logit of 20 at initialisation, and the scaled model 0.215; the trained model must learn to move between the two regimes.

Position extrapolation. Equation (6) shows a position shift is a rotation in each two-dimensional frequency plane, but the attention score is a bilinear form in the full embedding plus content, so relative-offset attention is available only if the learned projections align with the rotation structure. The authors test no sequence longer than training length, so the claimed extrapolation is untested; learned and sinusoidal encodings being equal at in-range lengths (row E) gives no information on it.

Low-resource and non-text domains. With 40,000 parsing sentences the model works, but the parser used a small hyperparameter search and structure-specific length settings. In domains with low signal-to-noise ratios and few observations, such as financial return series, the absence of a built-in sequential prior removes a form of regularisation that recurrence provides, so the same architecture may overfit; this is an inference and is not established by the paper.

Homogeneous benchmarks. The cost comparison depends on 2014 to 2016 era hardware and implementations; the ordering of architectures in cost may change with different accelerators or implementations, and the paper provides no sensitivity to that.

9 Further Literature

Table 6. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Bahdanau, Cho and Bengio (2015)	Soft alignment through weighted sums of encoder states	Preceded; additive attention is the alternative the paper compares with dot-product attention	Origin of the attention mechanism that the Transformer generalises to self-attention
Hochreiter and Schmidhuber (1997)	Gated recurrence for long-range dependencies	Preceded; the recurrent baseline family the paper replaces	Defines the sequential-computation bottleneck the paper targets
Gehring et al. (2017)	Convolutional sequence-to-sequence learning	Concurrent alternative with logarithmic or linear path length	The strongest non-recurrent competitor in Table 2
Parikh et al. (2016)	Decomposable attention between sentence pairs	Preceded; self-attention without recurrence in a narrower task	Closest prior use of attention without sequence-aligned recurrence
Shaw, Uszkoreit and Vaswani (2018)	Relative position representations in self-attention	Extended; replaces sinusoidal absolute encodings	Tests the paper's relative-position motivation directly
Devlin et al. (2019)	Bidirectional pretraining of Transformer encoders	Extended and applied	Shows the architecture transfers across tasks through pretraining
Kaplan et al. (2020)	Power-law scaling of loss with parameters, data and compute	Developed the scale property implied by the big model results	Provides the quantitative regularity the paper's two model sizes only hint at
Dosovitskiy et al. (2021)	Self-attention over image patches	Applied elsewhere	Tests whether the inductive-bias-free claim extends beyond text

Paper	Shared idea	Relationship	Why it matters
Zeng et al. (2023)	Simple linear models against Transformers for time series forecasting	Contradicts in the time-series setting	Shows attention's benefit is not automatic outside language
Gu, Kelly and Xiu (2020)	Machine learning for return prediction in empirical asset pricing	Applied in finance, without attention	Benchmark for transferring architectures to cross-sectional returns

Source: assessment by the reviewer. Entries other than the Vaswani paper are cited from the reviewer's knowledge and were not re-opened in this run.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No seed variance or significance testing. *Why it matters:* small BLEU differences drive the ablation conclusions. *Existing limitation:* one run per configuration. *Way forward:* multi-seed replication with paired tests.
- **Gap:** No test of long sequences or of positional extrapolation. *Why it matters:* the efficiency argument is conditional on $n < d$. *Existing limitation:* only sentence-length data. *Way forward:* length-stratified evaluation and synthetic copying tasks.
- **Gap:** No causal test of the interpretability claim. *Why it matters:* attention weights may not explain predictions. *Existing limitation:* three selected visualisations. *Way forward:* head ablation and intervention studies.

10.2 Literature-level gaps

- **Gap:** A theory of why softmax attention stacks generalise. *Why it matters:* without it, design is empirical. *Existing limitation:* expressivity results exist, generalisation results are weak. *Way forward:* sample-complexity analyses with realistic data distributions.
- **Gap:** Evidence on when attention beats simple baselines in noisy sequential data. *Why it matters:* finance and other low signal-to-noise domains. *Existing limitation:* benchmarks are dominated by language and vision. *Way forward:* standardised, compute-matched comparisons with proper out-of-sample inference.

10.3 Methodological gaps

- **Gap:** Cost reporting. *Why it matters:* peak-capacity proxies mislead. *Existing limitation:* no memory or inference cost. *Way forward:* report measured throughput, memory and energy by sequence length.
- **Gap:** Attribution between architecture and training recipe. *Why it matters:* claims about attention depend on it. *Existing limitation:* recipe changes in the same paper. *Way forward:* recipe-matched baselines.

11 Research Opportunities

Idea 1. Attention across assets as a stochastic discount factor

RESEARCH QUESTION. Does a self-attention layer applied across the cross-section of firms at each date, instead of a feed-forward network applied firm by firm, improve out-of-sample pricing of returns relative to the machine learning benchmarks in the empirical asset pricing literature?

MOTIVATION. Existing neural return predictors treat each firm independently conditional on characteristics, though returns co-move through common factors and peers; attention lets the weight on a firm's information depend on other firms in the same cross-section.

HYPOTHESIS. Cross-sectional attention increases out-of-sample R^2 and Sharpe ratios of long-short portfolios, mainly through weights on industry and size peers.

ECONOMIC MECHANISM. Information diffusion and lead-lag between related firms; attention weights as a learned, time-varying similarity matrix between firms.

LITERATURE CONNECTION. Gu, Kelly and Xiu (2020); Kelly, Malamud and Zhou (2024) on complexity in return prediction; Vaswani et al. (2017).

DATA. CRSP and Compustat characteristics; the public Jensen, Kelly and Pedersen factor and characteristic dataset as a freely available alternative.

VARIABLES. Monthly excess returns, standardised firm characteristics, industry and size codes.

METHOD. Transformer encoder over firms within a month, rolling or expanding training windows, loss on returns or portfolio utility, with the neural network of Gu, Kelly and Xiu as a baseline.

IDENTIFICATION STRATEGY. Strict out-of-sample splits, no look-ahead in standardisation, Diebold-Mariano tests across multiple seeds, and transaction-cost and capacity analysis.

EXPECTED CONTRIBUTION. A test of whether cross-sectional dependence, rather than nonlinearity alone, adds predictive value.

MAIN RISK. Low signal-to-noise ratio of monthly returns, so the attention layer overfits and gains vanish after costs.

Idea 2. Event-time positional encoding for irregular tick data

RESEARCH QUESTION. Does replacing integer position by continuous elapsed time or event count in the sinusoidal encoding, using the rotation property in Equation (6), improve forecasts of short-horizon price moves from limit order book events?

MOTIVATION. Trade and quote arrivals are irregular and clustered; integer position discards the information in inter-arrival times.

HYPOTHESIS. Time-aware encodings outperform index-based encodings at horizons where arrival intensity is informative.

ECONOMIC MECHANISM. Order flow clustering and price impact decay over event time, as in Hawkes-type models.

LITERATURE CONNECTION. Bacry, Mastromatteo and Muzy (2015) on Hawkes processes in finance; Vaswani et al. (2017); Shaw et al. (2018).

DATA. LOBSTER or other order book messages; the public FI-2010 benchmark as an accessible alternative.

VARIABLES. Mid-price changes, order imbalance, inter-arrival times, event types.

METHOD. Transformer with calendar-time, event-index and hybrid encodings; compute-matched comparison with a recurrent baseline.

IDENTIFICATION STRATEGY. Walk-forward evaluation across days and tickers, with placebo tests using shuffled arrival times.

EXPECTED CONTRIBUTION. Evidence on whether the positional design choice matters in irregular data, which the translation setting cannot reveal.

MAIN RISK. Gross predictability disappears after spreads and fees, so the result is statistical and not tradeable.

Idea 3. Softmax saturation under correlated embeddings

RESEARCH QUESTION. How does the variance of the pre-softmax logit in trained Transformers depart from d_k, and does a data-dependent temperature improve training stability?

MOTIVATION. Equation (2) assumes independence that fails after training.

HYPOTHESIS. Logit variance in trained models exceeds d_k in upper layers, so a learned per-head temperature stabilises optimisation.

ECONOMIC MECHANISM. Not applicable; the mechanism is statistical, concerning gradient flow through the softmax.

LITERATURE CONNECTION. Vaswani et al. (2017); work on pre-layer-norm stability (Xiong et al. 2020).

DATA. WMT or a small public language modelling corpus; trained checkpoints.

VARIABLES. Logit mean and variance by head and layer, entropy of attention rows, gradient norms.

METHOD. Instrumented training runs with multiple seeds; derivation of the variance under correlated components.

IDENTIFICATION STRATEGY. Controlled comparison of fixed and learned temperatures with matched compute.

EXPECTED CONTRIBUTION. An empirically grounded refinement of the paper's scaling argument.

MAIN RISK. Effects are small relative to seed noise.

Idea 4. Seed-level replication of the ablation table

RESEARCH QUESTION. Which differences in Table 3 survive repeated training with proper confidence intervals?

MOTIVATION. The head, key-size and width conclusions rest on single runs.

HYPOTHESIS. Differences below about 0.3 BLEU are not distinguishable from seed noise.

ECONOMIC MECHANISM. Not applicable.

LITERATURE CONNECTION. Vaswani et al. (2017), Table 3.

DATA. WMT 2014 English-German or the smaller IWSLT corpus if budget is limited.

VARIABLES. BLEU, perplexity, parameters, and training cost.

METHOD. Ten seeds per configuration, mixed-effects model of BLEU on configuration.

IDENTIFICATION STRATEGY. Randomised seeds and data order; paired bootstrap on the test set.

EXPECTED CONTRIBUTION. A calibrated version of the paper's most cited ablation.

MAIN RISK. Compute cost; the base model is about 3.3×10^{18} operations per run by the paper's own metric.

Idea 5. Compute-matched Transformer against simple models for return series

RESEARCH QUESTION. On daily and intraday return forecasting, do Transformers beat linear and recurrent models when the comparison is compute-matched and evaluated with proper multiple-testing control?

MOTIVATION. Attention is widely applied to financial series without comparison to simple baselines.

HYPOTHESIS. Gains over linear models are small or absent once costs and multiple testing are accounted for.

ECONOMIC MECHANISM. Weak predictability and nonstationarity limit the value of flexible architectures.

LITERATURE CONNECTION. Zeng et al. (2023); Gu, Kelly and Xiu (2020); Lim et al. (2021) on temporal fusion Transformers.

DATA. Daily equity and futures returns from public sources; intraday data where available.

VARIABLES. Lagged returns, volatility, volume, calendar variables.

METHOD. Walk-forward estimation, Superior Predictive Ability and Model Confidence Set tests.

IDENTIFICATION STRATEGY. Out-of-sample forecast comparison with pre-registered model grid.

EXPECTED CONTRIBUTION. A disciplined negative or positive result for a widespread practice.

MAIN RISK. Negative results are harder to publish and the grid can be argued to be unfair to the Transformer.

11.1 Ranking

Table 7. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Attention across assets	4	3	4	4	5	5	25
2. Event-time encoding	4	4	4	4	3	4	23
3. Softmax saturation	3	4	5	4	3	3	22
4. Seed replication	2	4	4	5	3	3	21
5. Compute-matched return forecasting	3	4	4	4	4	4	23

Source: reviewer's subjective scoring.

The top three are ideas 1, 2 and 5 in score order: idea 1 ranks first because it targets a central open question in empirical asset pricing with public data; ideas 2 and 5 tie at 23, and idea 2 is preferred for novelty and for using a feature of the paper (Equation (6)) that the original application never tested, while idea 5 is the most likely to produce a rigorous and publishable result about a common practice. Idea 3 follows at 22.

12 Learning Framework

12.1 Prerequisite concepts

- Encoder-decoder sequence-to-sequence models and autoregressive factorisation of $p(y_1, \dots, y_m \mid x)$.
- Recurrent networks, backpropagation through time and the vanishing gradient problem.
- Softmax, cross-entropy loss, perplexity and BLEU.
- Linear algebra of matrix products and orthogonal rotations; variance of sums of independent variables.
- Residual connections, layer normalisation, dropout and Adam.

12.2 Core concepts learned

- Attention as content-addressed averaging of values with weights from query-key similarity.
- Self-attention gives constant path length at quadratic cost in sequence length.
- Multi-head projections as parallel low-dimensional attention distributions.
- Position must be injected explicitly when the architecture is permutation-equivariant.

12.3 Important equations

Equation (1) defines the operation; Equation (2) justifies the scaling; Equation (3) defines the multi-head structure; Equation (6) shows why sinusoidal encodings make offsets linear; Equation (7) is the warm-up and decay schedule.

12.4 Methods learned

Masking to preserve autoregression, residual and layer-normalised sub-layers, label smoothing, checkpoint averaging, beam search with a length penalty, and cost accounting by training time times hardware capacity.

12.5 Questions to test understanding

COMPREHENSION

1. Explain in words what one row of the output of Equation (1) is.
2. Why does the decoder mask future positions, and where in the computation is the mask applied?
3. Why does the base model have $d_k = 64$ with eight heads?

ANALYSIS

1. Derive the variance of $q \cdot k$ and state the assumptions required.
2. For which n and d is self-attention cheaper per layer than recurrence?
3. Show that PE_{pos+k} is a linear function of PE_{pos} and identify the matrix.

RESEARCH

1. How would you test whether the gains come from attention or from the training recipe?
2. What changes when the input is a noisy financial series with few observations?
3. What evidence would show that a head performs a syntactic function?

12.6 Papers to read next

Bahdanau, Cho and Bengio (2015); Devlin et al. (2019); Kaplan et al. (2020); Shaw, Uszkoreit and Vaswani (2018); Gu, Kelly and Xiu (2020).

13 Conclusion

13.1 One-sentence thesis

A network built only from multi-head self-attention, feed-forward layers and residual connections can match and surpass recurrent and convolutional translation models at lower estimated training cost, so sequential recurrence is not required.

13.2 Three most important insights

- Removing sequential dependence in depth is what unlocks parallel training; the price is a cost quadratic in sequence length.
- The factor $1/\sqrt{d_k}$ is a variance normalisation: unscaled logits have variance d_k and saturate the softmax.
- Architecture claims need variance estimates; the paper's design is excellent, its statistics are thin.

13.3 Strongest evidence

The English-German result: 27.3 BLEU for the base model and 28.4 for the big model against 26.36 for the best earlier ensemble, with the base model costing 3.3×10^{18} operations.

13.4 Biggest weakness

Single-run results with no seed variance, no significance tests, and comparators taken from other papers.

13.5 Most important research gap

Whether and why attention generalises better than simple models in noisy, low-data sequential settings.

13.6 Best research opportunity

Cross-sectional attention for asset pricing with public data and strict out-of-sample evaluation.

13.7 Three papers to read next

Devlin et al. (2019); Kaplan et al. (2020); Kelly, Malamud and Zhou (2024).

14 References

Bacry, E., Mastromatteo, I., and Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1).

Bahdanau, D., Cho, K., and Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. *International Conference on Learning Representations*.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*.

Dosovitskiy, A., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations*.

Gehring, J., Auli, M., Grangier, D., Yarats, D., and Dauphin, Y. N. (2017). Convolutional sequence to sequence learning. arXiv:1705.03122.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223 to 2273.

Hochreiter, S., and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735 to 1780.

Kaplan, J., et al. (2020). Scaling laws for neural language models. arXiv:2001.08361.

Kelly, B., Malamud, S., and Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459 to 503.

Lim, B., Arik, S. O., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748 to 1764.

Parikh, A., Tackstrom, O., Das, D., and Uszkoreit, J. (2016). A decomposable attention model for natural language inference. *Empirical Methods in Natural Language Processing*.

Shaw, P., Uszkoreit, J., and Vaswani, A. (2018). Self-attention with relative position representations. *Proceedings of NAACL-HLT*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. Version read in full: arXiv:1706.03762, HTML rendering of the latest revision.

Xiong, R., et al. (2020). On layer normalization in the Transformer architecture. *International Conference on Machine Learning*.

Zeng, A., Chen, M., Zhang, L., and Xu, Q. (2023). Are Transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*.

Note: only the Vaswani et al. paper was opened and read in this run. All other works are cited from the reviewer's background knowledge and their bibliographic details were not re-verified; the dataset names (LOBSTER, FI-2010, Jensen, Kelly and Pedersen) are likewise named from background knowledge.